

APPLICATIONS

A VARIANCE COMPONENTS MODEL FOR STATISTICAL PROCESS CONTROL

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Summary: The classical Shewhart variables control chart plots a sequence of averages for which a simple model is assumed: a fixed "target" value and a component of random variation about that target. It is not always possible to explain the variation observed in industrial processes by such a simple model. We analyse the situation where replicate observations are obtained from each of a number of random samples taken daily (for example) from a process in which the variation in the process average is adequately described by a model allowing for substantial "day-to-day" ("between" days) variation and for "within" days variation in addition to the usual residual component of variance. We apply a two-way nested random effects analysis of variance model as the basis for estimating the components of variance which are then used to present a more appropriate control chart for daily averages. The median moving range, a robust estimator of variation, is suggested as a replacement for the more customary average moving range as an estimator for variation in the daily process average. A worked example is presented.

1. STATISTICAL MODELS FOR THE SHEWHART CHART

A production process is monitored by measuring a key response that conveys

information about the quality of the product. Statistical process control harnesses such information by means of a pair of control charts – one for variation and the other for the average – each with its own 3σ control limits. This methodology was originated by Walter A. Shewhart (1931) and further developed by later researchers (cf. Duncan, 1974, Ryan, 1989, Cryer and Ryan, 1990, Roes, Does and Schurink, 1993, and many others). Statistical quality control has become a vital part of modern process improvement techniques. In evaluating the SPC charts, the sampling procedure and the structure underlying the data are of cardinal importance: it contains fundamental information for understanding and explaining the observed variation in the response. Control limits based on an inappropriate model could be quite misleading.

The usual model for a Shewhart control chart for variables postulates that a quantitative observation consists of a constant target value plus random variation about that value. If a single measurement of the response variable made on the j -th item ($j = 1, \dots, J$), sampled on the i -th period of production ($i = 1, \dots, I$) is denoted by Y_{ij} , the Shewhart model is expressed as

$$(1) \quad Y_{ij} = \mu + e_{ij},$$

where μ is the fixed target value and e_{ij} denotes random variation about zero, usually assumed to be independently and normally distributed with unknown constant variance σ^2 . The average of each subgroup of J items sampled on the i -th period, $\bar{Y}_i$, is then plotted against i , to produce the well-known run chart. Supplemented with the control limits computed from (1) and the assumptions following it, the well-known Shewhart $\bar{X}$ and other control charts follow.

Models such as (1) are all important for statistical decisions since they determine the control limits uniquely. Proper model selection is the responsibility of the statistician and he or she has to verify that the chosen model provides an adequate description of the data produced by the process. If this is not the case little value can be attached to the conclusions based on

derivations from the assumed model. If a different model (for example an autoregressive time series model) provides a more satisfactory explanation for the behaviour of the response, decisions should be based on the statistical theory developed from that model (sometimes this is more easily said than done). The following description supports the example later selected to illustrate the situation where a model different from the simple classical model (1) provides a more satisfactory analysis.

The example deals with several subsamples taken from each of the items in a subgroup sampled on each period. To fix ideas, suppose that packages of yarn (each with thousands of metres of yarn on it) are produced by a continuous process and that J packages are sampled on day i ($i = 1, \dots, I$). Suppose further that K subsamples are taken from each package and tested for some physical property, for example *extension*. In the manufacturing environment where manual plotting of the data or rote application of standard statistical software takes place, it is quite likely that not even as simple a model as (1) will be recognized as underlying these procedures!

As introduced above, the variation observed could possibly be explained by several components such as a "between" days (day-to-day) component, a "within" days (package-to-package) component and a *residual* component. The latter may consist of several lurking components such as sampling variation as well as sample-to-sample (within packages) variation and experimental error. These sources of variation should be incorporated in a suitable model. In this paper the emphasis is on the proper analysis of a feasible model for the situation where nested sampling occurs. The results can readily be extended to more complicated analysis of variance type of models within this context (e.g. more factors with more involved nesting).

In the example selected for demonstration in this paper, a routine analysis of data where the within days variation was much smaller than the between days variation (unknown at the time), a standard pair of control limits turned out to be so narrow that practically every daily average was lying outside of the control limits (see figure 1) whereas it was known that

the process was producing acceptable product from the point of view of both the manufacturer and the customer. Until such time as the factors causing these fluctuations have been identified, there is no alternative but to consider the day-to-day variation as part of the common causes influencing the process. If all known sources of variation could be taken into account, the control limits will probably be wide enough to allow the manufacturer to continue a deliberate process improvement programme, rather than to keep stopping the process to hunt for special causes that can simply not be identified at that stage of development. In complicated continuous processes, in particular, it is never easy to identify variation uniquely with obvious sources such as daily setup problems, operator variation, shift-to-shift variation, identifiable batches of raw materials and so on. Setting up control limits based on a more realistic model will hopefully enable the improvement of the process to a stage where the day to day common causes will no longer be statistically significant. Finding and fitting the simplest relevant model should be part of the continuous improvement programme in the life of any process. The eventual aim is to have a process that produces data according to the model $Y_{ij} = \mu_0$, where the measurements take on a fixed preset value without any statistical variation!

2. A NESTED ANALYSIS OF VARIANCE MODEL FOR SPC

For the sampling procedure described above, a typical observation (the k -th of K) from the j -th package ($j = 1, \dots, J$) sampled on day i ($i = 1, \dots, I$) is denoted by Y_{ijk} . The variation observed in such a response could be explained in terms of a classical analysis of variance model providing for an overall target value, for main and possibly interaction effects as well as some unexplained residual variation. Both the "days" (D) and the "packages" (P) factors may be considered as random factors with "packages" hierarchically nested within "days". Thus the model accepted here is a two-way nested ANOVA (cf. Neter, Wasserman and Kutner, 1985):

$$(2) \quad Y_{ijk} = \mu + a_i^D + a_{j(i)}^{P(D)} + e_{k(ij)}$$

This model explains each data point as being additively influenced by three random effects (*days*, *packages-within days* and *residual*). The variance of each data point therefore consists of three components corresponding to each of these random effects. The notation in (2) indicates that K does not have to be constant but could depend on i and j (where j itself may depend upon i). For the sake of simplicity we shall assume them to be constant, viz. $i = 1, \dots, I$, $j = 1, \dots, J$ and $k = 1, \dots, K$. The variance components of *days*, *packages* and *residual* effects are denoted by $\sigma_D^2 := \text{Var}[a_i^D]$, $\sigma_{P(D)}^2 := \text{Var}[a_{j(i)}^{P(D)}]$ and $\sigma_e^2 := \text{Var}[e_{ijk}]$ respectively. As part of the analysis of variance model these effects are assumed to be independent hence

$$(3) \quad \text{Var}[Y_{ijk}] = \sigma_D^2 + \sigma_{P(D)}^2 + \sigma_e^2$$

The estimation of the variance components and its application to the construction of control charts will be considered in the next section.

3. ESTIMATING THE VARIANCE COMPONENTS

The variable to be charted against i is the daily average measured response, $\bar{Y}_{i..} := \sum_{j,k} Y_{ijk} / JK$.

From (2) and its assumptions it follows that $E[\bar{Y}_{i..}] = \mu$, which may be estimated from the data if a target value is not known. If $\bar{Y}_{i..}$ is the main quantity of interest, it follows from (3) that its variance may be partitioned into three components as follows:

$$(4) \quad \sigma_{\bar{Y}_{i..}}^2 := \text{Var}[\bar{Y}_{i..}] = \sigma_D^2 + \sigma_{P(D)}^2 / J + \sigma_e^2 / JK.$$

The analysis of variance for the model in (2) is presented in table 1 (see Scheffé, 1959, or Neter, Wasserman and Kutner, 1985, pp. 973 and 977).

TABLE 1. ANOVA table for nested two-factor random effects model (P nested within D)

Source of variation	SSQ	DOF	MSQ	E[MSQ]
Factor D	$SSD = JK\sum(\bar{Y}_{i-} - \bar{Y}_{-})^2$	$I-1$	MSD	$\sigma_e^2 + JK\sigma_D^2 + K\sigma_{P(D)}^2$
Factor P(within D)	$SSP(D) = K\sum\sum(\bar{Y}_{ij-} - \bar{Y}_{i-})^2$	$I(J-1)$	$MSP(D)$	$\sigma_e^2 + K\sigma_{P(D)}^2$
Residual	$SSR = \sum\sum\sum(Y_{ijk} - \bar{Y}_{ij-})^2$	$IJ(K-1)$	MSR	σ_e^2
Total	$SST = \sum\sum\sum(Y_{ijk} - \bar{Y}_{-})^2$	$IJK-1$		

The mean sums of squares are denoted by MSD , $MSP(D)$ and MSR and are obtained by dividing the corresponding sums of squares by their degrees of freedom (DOF). The components of variance may be estimated from an existing data set using table 1, as follows:

$$(5) \quad \hat{\sigma}_D^2 = (MSD - MSP(D)) / JK,$$

$$\hat{\sigma}_{P(D)}^2 = (MSP(D) - MSR) / K,$$

$$\hat{\sigma}_e^2 = MSR.$$

These are known as the ANOVA estimators of the variance components (cf. Searle, Casella and McCullough, 1992) and are presented for estimating the contribution of each of the effects towards the total variation for a given response variable. It is well known that this method of estimation may result in difficulties. For example, it could yield *negative* estimates of the variance components – which seems like a contradiction in terms. For a full discussion of resolving such a situation the reader is referred to Searle, Casella and McCullough, 1992.

We are now in a position to evaluate the contribution of each factor in the ANOVA model towards the variation of an individual observation or, since our main interest here is in the day-to-day averages, towards the variation in $\bar{Y}_{i-}$. From (4) it follows that

$$(6) \quad s_{\bar{Y}_{i..}}^2 \equiv \hat{\sigma}_{\bar{Y}_{i..}}^2 = \hat{\sigma}_D^2 + \hat{\sigma}_{P(D)}^2 / J + \hat{\sigma}_e^2 / JK.$$

Since the variances do not depend on i we drop the index from the following formulas for convenience. From (5) and table 1 we see that (6) simplifies to the intuitive estimator:

$$(7) \quad s_{\bar{Y}}^2 = MSD / JK = \frac{1}{I-1} \sum_{i=1}^I (\bar{Y}_{i..} - \bar{Y}_{..})^2.$$

As is evident from (6) all the components of variance are taken into account when estimating the variation of $\bar{Y}_{i..}$ and not only the residual variance as would be the case in the classical Shewhart model. The contributions of the three components of variance towards the total variation and their estimators, derived from (5) and (7), are given in table 2.

TABLE 2. Estimated contributions of the *main* factor, D , the *nested* factor, $P(D)$, and the *residual* component towards variation in the average over the levels of D

Factor →	D	$P(D)$	Residual	Total
Contribution - from (6)	σ_D^2	$\sigma_{P(D)}^2 / J$	σ_e^2 / JK	$\sigma_{\bar{Y}}^2$
Estimated proportion	$\hat{\sigma}_D^2 / s_{\bar{Y}}^2$	$\hat{\sigma}_{P(D)}^2 / Js_{\bar{Y}}^2$	$\hat{\sigma}_e^2 / JKs_{\bar{Y}}^2$	1
From (5)	$= 1 - \frac{MSP(D)}{MSD}$	$= \frac{MSP(D) - MSR}{MSD}$	$= \frac{MSR}{MSD}$	

4. SELECTING CONTROL LIMITS FOR THE AVERAGES CHART

The components of variance model leads to the estimation of the standard deviation of $\bar{Y}_{i..}$ according to (7) and this implies that control limits for $\bar{Y}_{i..}$ - plotted against its (*daily*) index, i - may be obtained in principle as if the daily averages were considered to be individual observations on an *individuals* chart (an X chart), their sample standard deviation being equal to the "between days" standard deviation.

These control limits for the daily average, $\bar{Y}_{i..}$, may be set up as $\mu_0 \pm 3\sigma_{\bar{Y}}$ where μ_0 is a selected target value. If the target value is to be data-driven it could be estimated from the review data set as $\hat{\mu} = \frac{1}{I} \sum_i \bar{Y}_{i..}$ or as $\hat{\mu} = \text{Median} \{ \bar{Y}_{i..}, i = 1, \dots, I \}$. Cryer and Ryan (1990) advocate the overall standard error for all individuals (here computed over all available daily averages) after correcting with the bias correcting constant $c_4 = c_4(I)$ as the preferred method of establishing control limits for an *individuals chart*. Hence the control limits for $\bar{Y}_{i..}$ are

$$(8) \quad \mu_0 \pm 3s_{\bar{Y}} / c_4.$$

If the process under investigation is "in control" there could be little argument against using the sample standard deviation estimated from the ANOVA table as measure of variation. It is the most efficient estimator for $\sigma_{\bar{Y}}$ under those assumptions. Nevertheless, efficiency is not the only yardstick by which to select an estimator. It is of little value to use one which is efficient in a given model under circumstances where that model is not applicable. We thus digress momentarily to motivate an alternate estimator for variation following the reduction of the original model to one of the averages as if they were individual observations. We start by asking what precisely does it mean when we say a process is "in control"? We can do no better at explaining this than by quoting Shewhart (1931):

"...a phenomenon will be said to be controlled when, through the use of past experience, we can predict, at least within limits, how the phenomenon may be expected to vary in future. Here it is understood that prediction within limits means that we can state, at least approximately, the probability that the observed phenomenon will fall within the given limits".

How does one use "past experience" to be in a position to claim (with associated probability) that a process response is within predicted limits? This is only possible by using past data as guidance on the selection of a model and to derive the prediction formula and the associated probability for predictions from that model. This confirms our already stated contention that the choice of model for process control is vital and without it no serious statistical analysis can be undertaken.

The selection of estimators for the parameters in the model is as important as the choice of model itself. Notwithstanding the arguments put forward by Cryer and Ryan (1990), for the model for *individual measurements* (also studied by Roes, Does and Schurink, 1993), there remains some argument as to which estimator to use as a measure of variation. The purists, believing in the truth of the "*in control*" state of the process on which control limits are to be based, will stay with (8) but current practice seems to prefer the average moving range estimator for neighbouring pairs, $\overline{MR}_2$, originally introduced by Duncan, 1974 (cf. Wheeler, 1990, 1995 and Wheeler and Chambers, 1992) and formally defined as

$$(9a) \quad \overline{MR}_2 \equiv \frac{1}{I-1} \sum_{i=2}^I |\bar{Y}_i - \bar{Y}_{i-1}| / d_2$$

We refer to this as the "moving range" estimator of the standard error, $\sigma_{\bar{Y}}$. The constant $d_2 = d_2(2) = 1.1284$ is the bias-correcting factor for the range estimator of the standard deviation from a $N(\mu, 1)$ sample of size 2. The control limits based on (9a) are

$$(9) \quad \mu_0 \pm 3\overline{MR}_2$$

Both $s_{\bar{Y}}/c_4$, and $\overline{MR}_2$ are unbiased for the standard deviation, $s_{\bar{Y}}/c_4$ being most efficient (i.e. smaller variance) for the SD of the normal distribution. Even though $\overline{MR}_2$ is not as efficient as the standard deviation when the assumption of "*in control*" is true - it is a pragmatic choice

because the data on which the control limits are to be based are often not in control even when the process at that point is believed to be in control. The future control chart needs protection against possible violations of that kind! This is achieved to an extent by $\overline{MR}_2$ since $\frac{s}{\bar{Y}}$ is quite non-robust, being influenced severely by even a single outlier – it has an unbounded influence function and moreover, a poor breakdown point. Wheeler, (1990, p. 51) says that $\overline{MR}_2$ is “*much more insensitive to lack of control*”. Since $\overline{MR}_2$ only uses information from temporally contiguous pairs of data points it is influenced to a lesser extent by overall out of control processes. However, the moving range is by no means the most outlier resistant estimator either. It suffers from the ordinary deficiencies inherited from its ancestry, being range-based as well as an average. A single outlying jump between a neighbouring pair of averages will influence it as seriously as it would the standard deviation. Such an outlying jump, for example, will induce an unduly increased estimate of the process variation. This in turn will lead to control limits that are too wide and hence out of control points on a future chart will go undetected. With these facts in mind, we introduce an alternative set of control limits derived from a robustification of $\overline{MR}_2$ by taking the median rather than the average of the moving subgroup ranges, as follows:

$$(10) \quad \mu_0 \pm 3 \text{Med}R_2, \quad \text{where}$$

$$(10a) \quad \text{Med}R_2 = \text{Median}\{|\bar{Y}_i - \bar{Y}_{i-1}| / d_4, \quad i = 2; \dots; I\}.$$

The control limits (10) are based on the median of the moving ranges for neighbouring pairs, to which we shall refer as the “median moving range”. Its bias correcting factor when estimating σ , is $d_4 = d_4(2) = 0.9539$, being the median of the range of a sample of size 2 from a $N(\mu; 1)$ distribution (Harter, 1960, table 1, lists quantiles of the sample ranges from a $N(\mu; 1)$ distribution). It retains the favourable property of using neighbouring pairs of observations to measure variation, yet it is resistant against outlying pairs by being a median.

If there is a consideration in favour of robust estimation, it is advisable

not only to use a robust estimator for scale but also a robust estimator for location if the target value has to be estimated. Hence the median, $\mu_0 = \text{Median} \{ \bar{Y}_i, i = 1, \dots, I \}$, or the more efficient Huber 1.5 estimator (cf. Hogg, 1979) would be a good substitute. The latter has the disadvantage of being more cumbersome to compute. According to Roes, Does and Schurink (1993) the median as location estimator gives no noticeable loss in efficiency compared to the average.

Apart from possibly unexpected abrupt changes in the data that will affect $s_{\bar{Y}}$ more dramatically than it will $\overline{MR}_2$ or $MedR_2$ these estimators will, in general, give a more relevant reflection of the variation reflected by the bulk of the data used for setting up control limits for a process. Similar remarks could be made in support of either $\overline{MR}_2$ or $MedR_2$ as estimator of choice for day-to-day variation if the data are persistently drifting away from a central target, following a trend. In such cases, regression models should be considered as alternatives in which case classical or robust regression estimators could be used. Under these conditions $s_{\bar{Y}}$ used in conjunction with an unsuitable model will overestimate and provide over-conservative (wider) control limits. The same applies to data showing cyclic behaviour for which a spectral control chart should be used (cf. Spurrier & Thombs, 1990).

5. THE CORRESPONDING RANGE CHART

The variance components model (2) implies that the daily averages should formally be treated as if they were individual observations. When plotting a regular control chart for individuals it is argued by Roes, Does and Schurink (1993) that the moving range chart for monitoring variation does not contribute worthwhile information not already reflected in the individuals control chart for averages – a judgement with which we agree. Hence, also in the components of variance model, the control chart for averages reflects all the available information concerning “between” days variation. However, when monitoring the “within” days process variation the sample range is

available for each day (which is not the case in a standard individuals chart) and a classical range chart of R_i versus i should be constructed. Since the sample range is location invariant (i.e. it does not depend on the level of the data for a given day, but only on the variation in the data for that day) which by assumption remains the same from day to day, the usual

("within") range chart with target value $\bar{R} = \sum_{i=1}^I R_i / I$ and control limits

$D_3(J)\bar{R}$ (lower) and $D_4(J)\bar{R}$ (upper), where D_3 and D_4 are the classical range control chart parameters (cf. Ryan, 1989, p. 143 and table E - Control chart constants, p. 434) is recommended as part of the proper practice of SPC when using the model (2).

As far as the range chart is concerned, suppose we use $\bar{R} = \sum_{i=1}^I R_i / I$ (where R_i is the range of subgroup i) as target value and control limits based on I subgroups of size J where each observation in the subgroup (and in our model this may be the average of K replicates) is normally distributed with standard deviation σ . The influence of the subgroup size on the variability of the range control chart is derived by considering the standard deviation of the estimated target value and the SD of the width of the control region (here called the *precision*).

The control limits are $\bar{R} \pm 3(d_3/d_2)\bar{R}$, (Ryan, 1989, p. 143). Here $d_2(J)$ and $d_3(J)$ are the usual control chart constants. The SD of $\bar{R}$ is $\sigma_{\bar{R}} = d_3(J) \cdot \sigma / \sqrt{I}$ and that of the precision is $6(d_3/d_2) \sigma_{\bar{R}}$. Hence, dropping the constant $6\sigma/\sqrt{I}$, the SD of the precision is proportional to d_3^2/d_2 . For fixed I these scale factors decrease with J as presented in table 3.

TABLE 3. Changes in the SD of range and the SD of precision with subgroup size

J	2	3	4	5	6	7	8
$d_3(J)$	0.8525	0.8884	0.8798	0.8641	0.8480	0.8332	0.8198
$d_3^2(J) / d_2(J)$	0.6441	0.4663	0.3760	0.3210	0.2838	0.2567	0.2361

Effectively this means, for example, that increasing the subgroup size from $J = 4$ to $J = 8$, the variability of the estimated target value (measured by its SD) is reduced from 0.88σ to 0.82σ (disregarding the factor $1/\sqrt{I}$), that is by about 7%. For the precision the corresponding variability is reduced from 0.38σ to 0.24σ , a relative reduction of approximately 37%.

6. COMPONENTS OF VARIANCE FOR A SINGLE REPLICATE

It sometimes happens that only a single measurement ($K = 1$) is available per package or that the laboratory reports only the average, even though K observations were done per package (real life!). In this case two indices are required to identify the observation, viz. $Y_{ij} \equiv Y_{ij1}$ and the model corresponding to (2) becomes: $Y_{ij1} = \mu + a_i^D + a_{j(i)}^{P(D)} + e_{ij1}$.

Since the last two terms are confounded (having non-distinguishable indices), this model does not allow separate estimation of $\sigma_{P(D)}$ and σ_e and the model may equally well be written

$$(11) \quad Y_{ij} = \mu + a_i^D + e_{ij},$$

where $e_{ij} := a_{j(i)}^{P(D)} + e_{ij1}$ denotes the residual component which now includes both package-to-package variation and experimental error. Note that $\text{Var}[e_{ij}] = \sigma_{P(D)}^2 + \sigma_e^2$, that the model is no longer nested and that it reverts to a one-factor (random effects) analysis of variance model. The ANOVA for model (11) is presented in table 4 (see Scheffé, 1959, p. 226, for the results in the expected mean squares column).

TABLE 4. ANOVA table for a single factor random effects model

Source of variation	SSQ	DOF	MSQ	$E[MSQ]$
Factor D	$SSD = J \sum (\bar{Y}_{i.} - \bar{Y}_{..})^2$	$I - 1$	MSD	$\sigma_e^2 + J\sigma_D^2$
Residual	$SSR = \sum \sum (Y_{ij} - \bar{Y}_{i.})^2$	$I(J - 1)$	MSR	σ_e^2
Total	$SST = \sum \sum (Y_{ij} - \bar{Y}_{..})^2$	$IJ - 1$		

As before, we require a chart, with control limits, for $\bar{Y}_{i.}$ against i . From (11) and table 4 we see that its variance, $\sigma_{\bar{Y}_{i.}}^2$, is (dropping the index i as before)

$$\text{Var}[\bar{Y}_{i.}] = \sigma_{\bar{Y}}^2 = \sigma_D^2 + \sigma_e^2/J = E[\text{MSD}/J]$$

from which it follows that an unbiased estimator for $\sigma_{\bar{Y}_{i.}}^2$ is

$$(12) \quad s_{\bar{Y}}^2 = \text{MSD}/J = \sum (\bar{Y}_{i.} - \bar{Y}_{..})^2 / (I - 1).$$

Note that the proportional contribution of factor D to the variance of $\bar{Y}_{i.}$ in the model (11), similar to that derived in table 2, is $\hat{\sigma}_D^2 / s_{\bar{Y}}^2 = 1 - \frac{\text{MSR}}{\text{MSD}}$

whereas the residual component contributes a proportion $\frac{\sigma_e^2/J}{s_{\bar{Y}}^2} = \frac{\text{MSR}}{\text{MSD}}$.

Using (12) in the same way and with similar comments to those discussed previously, the control limits on $\bar{Y}_{i.}$ may be computed as $\mu_0 \pm 3s_{\bar{Y}}/c_4(I)$

or as $\mu_0 \pm 3\overline{MR}_2$ but our recommendation, as before, is to use

$$(13) \quad \mu_0 \pm 3\text{Med}R_2,$$

where μ_0 is a selected target value or an estimate as discussed before. For this model, as in (8), $s_{\bar{Y}}$ in (12) is the sample SD of the daily averages in the review data set used for estimating the control limits. Thus there is no difference in principle in the analysis of a single replicate and in that of multiply replicated data.

7. MODUS OPERANDI

The procedure for evaluating the appropriateness of a components of variance model and its subsequent application for SPC is summarized in the

following sequence of steps.

- Prepare a Shewhart standard deviation or range control chart to study the process variation.
- If the process variation appears to be stable, estimate a target value by computing both the average and the median (or another robust estimator of location) of the daily averages. If these differ considerably from one another or from a known target value, outlier detection methods should be applied (cf. Iglewicz and Hoaglin, 1993) and the review data set should be cleaned. Then plot and inspect the *run chart* for daily averages.
- Identify possible data patterns such as substantial shocks, trends or persistent cycles and consider an appropriate model in preference to a variance components or a simple Shewhart model. The following steps are followed to investigate the components of variance model.
- Compute the analysis of variance as in table 1 and use the nested random effects F-test (use $MSD/MSP(D)$ upon $I - 1$ and $I(J - 1)$ DOF) to test for significance in daily averages.
- Use the ANOVA table and table 2 to estimate the proportional contribution of each of the components of variance to the variation occurring in the daily averages. If the "between days" component is large the components of variance model (2) is probably applicable.
- Next estimate the "between days" sampling standard error by using $s_{\bar{Y}}/c_4(I)$, $\overline{MR}_2$ and $MedR_2$ for comparative purposes. If they are practically similar, use (8) as control limits. If they differ perceptibly, use (10). If there are substantial differences between them it is virtually sure that a problem of one type or another exists in the data. Such

problems have to be identified (not necessarily an easy task) before further progress can be expected on the path of continuous quality improvement. This is in line with a similar recommendation made by Cryer and Ryan (1990) and in line with the principles of sound data analysis.

8. APPLICATION

The (new in 1995) Ultracord plant at the Bellville works of *S.A. Nylon Spinners (Pty.) Ltd.* wants to set up SPC procedures for the continuous process 1400f204 T1837, by daily sampling $J = 8$ packages of spun yarn per machine. Various physical properties of the yarn are replicated in the laboratory by analysing five samples of yarn per package ($K = 5$). The yarn property of *extension* (the percentage increase in length before breaking) is selected for illustrative purposes. Packages with yarn sampled for the first $I = 15$ days of January 1995 are selected as review data set to determine control limits for future use in monitoring the process.

If the average of each set of $K = 5$ replicates is taken as representative for each package, then with $J = 8$ packages sampled per day, the daily ranges are computed and the average range over $I = 15$ days is found to be $\bar{R} = 1.44$. A range control chart is then set up with target value of 1.44 and control limits $0.1362^*1.44 = 0.20$ and $1.8638^*1.44 = 2.68$ using the control chart constants $D_3(8)$ and $D_4(8)$. The daily ranges plot well within these limits (chart not shown here) and hence process *variation* is accepted to be stable.

The overall average for *extension* for the data set is $\bar{Y}_{..} = \hat{\mu} = 20.96$ and the median of the daily averages is $\tilde{\mu} = 20.98$. These estimates are so close together that it does not really matter which one is chosen as target value. Their closeness points to a symmetrical distribution for daily averages. If the control limits for daily averages are computed in the usual way from the Shewhart model (1), using the average sample range to estimate standard error, the control limits are $20.96 \pm 0.3725^*1.44 = 20.96 \pm 0.54$ using the

control chart constant $A_2(8) = 0.3725$. The estimate of the standard error based on the adjusted average of the subgroup ranges $\left[\frac{1.44}{d_2(8)\sqrt{8}} = 0.18 \right]$ is so small (and retrospectively we can say this is due to model inadequacy) and hence the control limits so narrow, that most future data points will be declared "out of control". It is this phenomenon that stimulated this research. The run chart is presented in figure 1 in which the values computed above are indicated as the inner pair of control limits.

The components of variance model (2) is thus accepted for evaluation and the analysis of variance results are presented in table 5.

TABLE 5. ANOVA table for *Extension* measurements on *SANS Ultracord* data for 1400f204 T1837 yarn

Source of variation	Sum of squares	DOF	Mean square	F	P
Days	395.4342	14	28.2453	22.36	0.0000
Packages (within days)	132.6570	105	1.2634	1.55	0.0011
Residual	390.6720	480	0.8139		

The F test shows highly significant differences in daily averages and the other information in table 5 is used to estimate the relative importance of the sources of variation as in table 2. The components of variance, using (5), are $\hat{\sigma}_D^2 = 0.6745$, $\hat{\sigma}_{P(D)}^2 = 0.0899$ and $\hat{\sigma}_e^2 = 0.8139$. Hence the proportion of variation for daily averages accounted for by *package-to-package* (within days) variance is (from tables 2 and 5) $\frac{1.2634 - 0.8139}{28.2453} = 1.6\%$. The *residual* variance accounts for $0.8139/1.2634 = 2.9\%$ and the balance of 95.5% is due to *day-to-day* variation and hence this data set is clearly favouring the components of variance model.

We now compute the control limits based on the components of variance model. Even though we recommend using $MedR_2$ and (10) for the control limits in general, we nevertheless compare the three estimates for variation

in daily average. These are $s_{\bar{Y}}/c_4 = 0.84/0.9823 \doteq 0.86$ (from (7) and table 5), $\overline{MR}_2 = 0.64$ and $MedR_2 = 0.45$. The large differences in estimates of the standard error are in agreement with the large between days component of variance.

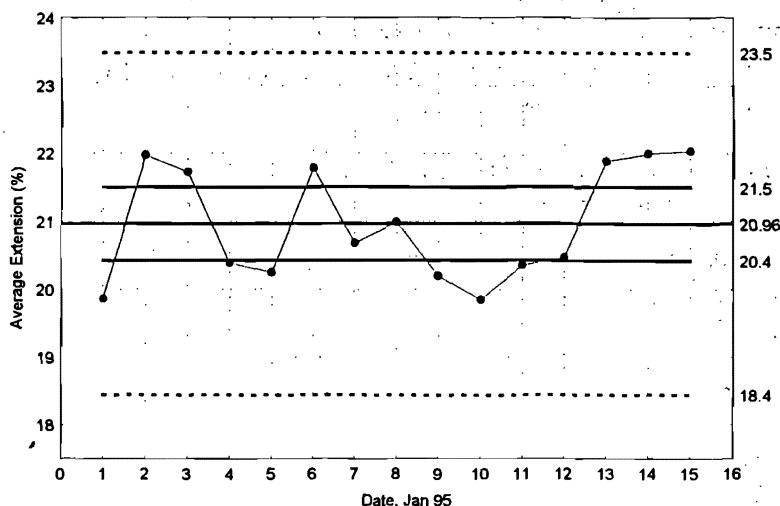


FIGURE 1. Control chart for daily average *Extension* of high tenacity yarn for $I = 15$ subgroups of size $J = 8$. Inner control limits (solid lines) based on the average of the subgroup ranges ($\bar{R} = 1.44$) and the outer limits (dotted lines) on the $SD(s_{\bar{Y}} = 0.84)$ of the daily averages

The control limits based on $s_{\bar{Y}}$ is 20.96 ± 2.57 (see figure 1) whereas for $\overline{MR}_2$ they are 20.96 ± 1.92 (see figure 2, for illustrative purposes only). For $MedR_2$ the control limits are 20.96 ± 1.35 and these are the future control limits for the chart - see figure 2. The chart does not reveal any

extraordinary responses that could be linked to special causes (which would imply outright rejection or at least testing for outliers). Apart from the daily averages, figure 2 also shows the positions for each set of 8 individual observations underlying these averages. Normally these would not be shown on the control chart.

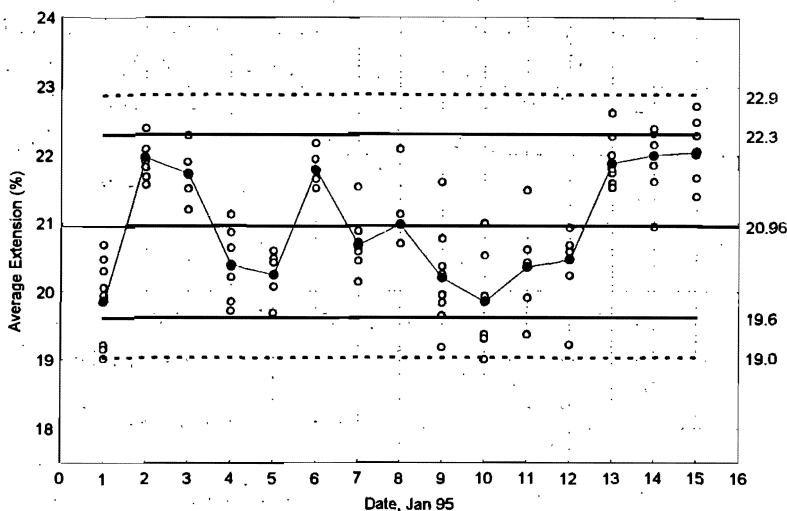


FIGURE 2. Control chart for daily average *Extension* of high tenacity yarn (filled circles) with control limits based on $MedR_2$ (solid lines) and on $\overline{MR}_2$ (dotted lines). Data for individual packages are superimposed as open circles.

9. RECOMMENDATION

The choice of model underlying a process should be made with circumspection. Specific processes may be well described by the simple Shewhart model but for others more complicated structure may be required. It is the task of the statistician to find an appropriate model describing the process data at hand and to base conclusions on that model. In this paper we study a situation where an analysis of variance model with two hierarchical random effects factors can be used to describe the process response. If the "within"

variance component is small relative to the "between" component, it is suggested that the control limits for the (daily) averages of the process should be based on the median pairwise moving range, $MedR_2$, as the estimator of choice for the process variation. Under these circumstances it was argued (on the basis of its robustness properties) that the median moving range statistic should in general provide narrower control limits than those provided by the average moving range statistic, which in turn should provide less conservative bounds than the standard deviation estimator. The "within" variation of the process should be monitored in the classical way with the use of a regular standard deviation or range control chart.

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